Verified, but What?
Certificate Scope in a Finite Agent Pipeline

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Abstract

An exact local proof need not establish the task a user originally requested. We examine this boundary in a finite translator–solver–reviewer pipeline with strict Boolean contracts and independently checked exhaustive certificates. Honest translation is separated from planted handoff faults. For fixed original tasks, altered candidates and valid local evidence, a model reviewer sees an ordinary PASS heading, a neutral heading or explicit candidate-only scope wording. Every condition retains the original task and correspondence-checking instruction. After a passing development pilot, the frozen Qwen3.5-122B-A10B evaluation covered 100 new originals and 260 scheduled roles. All three presentations returned correct original counterexamples on all 40 planted faults and correct SAFE answers on all 20 equivalent controls. Primary and secondary paired false-SAFE differences were zero on 40 complete pairs each; degenerate bootstrap intervals do not establish equivalence, and the bounded-iid interval was approximately [-0.4295,0.4295]. Natural translation preserved both open contract functions in 37/37 valid translations, while three malformed translations and one malformed review reduced useful completion to 36/40. The main required 259 physical attempts, including two identical technical retries with unknown failed-attempt charges retained in the budget. No false guarantee was observed. The result is a reproducible null under an explicit, small-task boundary condition, not a general mathematical equivalence criterion, demonstrated scope-label benefit or model-safety guarantee.

1 Introduction

A correct proof answers the statement it proves, not automatically the task a user intended. An agent team can therefore produce locally impeccable evidence and still answer the wrong question. The scientifically relevant issue here is not whether a verifier can validate a finite proof, but how a downstream agent uses that proof when the original task remains visible.

We study a deliberately small translator–solver–reviewer pipeline. The translator and reviewer are separate model contexts; the solver and certificate checker are local code. We separate honest translation drift from deliberately inserted handoff faults. On fixed original tasks, candidates and valid local certificates, we compare an ordinary PASS heading (B), a neutral heading (N), and an explicit candidate-only scope statement (S). Our contributions are a fully checkable finite-contract measurement setup, a paired presentation comparison, and an auditable accounting of correctness, useful completion and uncertainty. Neither a new general equivalence criterion nor a novel guarantee from an oracle gate is claimed. The result-specific conclusion is stated in Section 6.

2 Related Work

Li et al.'s SemantEq [1] explains why unrestricted bidirectional implication can equate unrelated closed theorems: both may be independently provable. Its proof-term policy addresses faithful formal statement comparison. Our setting is narrower. Atom meanings and inputs are fixed, and we compare the admissibility and obligation functions separately. This is a contract-preservation convention, not a solution to faithful mathematical autoformalization.

Mei and Douglas [2] combine semantic review with mechanical numerical and vacuity checks for agentic autoformalization. The insufficiency of checking only a proof is established prior work. Our comparison instead holds the actual local evidence fixed and varies how its scope is presented to the reviewer. We do not reproduce either paper's benchmark, quote their findings as our measurements, or claim priority for statement/proof mismatch.

3 Methods

3.1 Contracts and exact ground truth

For four Boolean inputs $x = (p, q, r, s)$ with fixed meanings, a contract is a pair (C, Q). Its original claim is

$$\forall x \in \{0,1\}^4 : \; C(x) \Rightarrow Q(x).$$

SAFE means this claim is true. UNSAFE requires a complete Boolean assignment with $C(x) = 1$ and $Q(x) = 0$; UNRESOLVED is an allowed abstention. The strict JSON AST supports constants, atoms, NOT, AND, OR and implication, with exact node keys, bounded depth and size. Arbitrary code and `eval` are not used.

The original AST is rendered in an unambiguous controlled notation. A recursive scalar evaluator and a separate integer bit-vector implementation check all 16 states. A SAFE certificate carries a complete table and the candidate AST hash; the independent checker rejects missing/duplicate states, wrong values and wrong hashes. UNSAFE witnesses are independently checked. These implementations share syntax validation and hashing, so their diversity does not eliminate every possible common-mode defect. Empty-domain/vacuity checks are unit tests, not a main fault family.

Open-function drift means $C \not\equiv C'$ or $Q \not\equiv Q'$. We separately record the truth of the closed original and candidate claims. Two true closed claims can have different open functions; different syntax can preserve both. A changed intermediate contract need not cause a wrong final answer. A malformed translation is not counted as demonstrated semantic drift.

3.2 Populations and paired intervention

The preplanned evaluation has 100 new originals. Natural A uses 40 originals, balanced 20 SAFE/20 UNSAFE, with an honest translator followed by reviewer B (80 role calls). Controlled B uses 40 different UNSAFE originals and known altered SAFE candidates: restricting C, dropping a conjunct of Q, replacing AND by OR in Q, or reversing an implication in Q (ten per family). Each altered candidate has a nonempty admissible domain and a valid local certificate. Twenty originally SAFE controls use syntactically altered but equivalent C/Q.

Each controlled or equivalent original is reviewed under B, N and S in separate fresh contexts (180 role calls). All arms see the original task, candidate AST, certificate and hash. The common instruction explicitly asks the reviewer to answer the original and check correspondence. S receives no extra reference AST or counterexample. The headings are approximately character-matched, not claimed to be tokenizer-matched; the intervention includes semantic scope wording, not merely typography. Case and condition orders are seeded and frozen. There is no no-certificate arm or budget-matched single-agent comparison.

3.3 Pilot, model and prospective controls

The pilot schedules eight natural, eight planted and four equivalent originals: 52 role calls. At least 47/52 responses must parse and at least 10/12 clean originals must be correct; an equivalent original must be correct in all three arms. A null fault result does not fail this gate. Main requires a passing audited pilot, prospective freeze and a separate source-bound authorization.

The fresh seed is 2026110816. Fourteen released/inspected engineering bundles are excluded. Original/candidate paired C/Q truth tables and original atom-erased AST skeletons do not overlap development and evaluation; actual pilot translations are checked too. Higher-level mutation families remain shared, so this is not an unseen-family benchmark. Selection uses exact mathematics, not model outcomes.

The experimental subject is `qwen/qwen3.5-122b-a10b` through OpenRouter's fixed Alibaba route [3, 4]. Temperature is 0.6, top-p 0.95, reasoning enabled and its text excluded, requested output 8192 tokens, and no

fallback provider or JSON response mode. Visible answers and usage are retained; hidden reasoning text is not. The public model alias is not an immutable weight pin, and provider quantization is unknown.

The study allocation is USD20. Each dispatch first reserves USD0.61341696, covering the full advertised 262144-token context separately at input/output tariffs USD0.26/2.08 per million. Actual total input plus output is checked against the advertised context. Native requested output limits are not assumed to cap all thinking tokens. Complete valid billing settles the larger of reported cost and the no-cache token-price bound; unknown charges retain the full reserve. The upfront-full-campaign-fit rule was prospectively relaxed, not the per-request hard wallet. Completion is therefore not guaranteed; a budget stop leaves scheduled cases unassessed rather than shrinking the sample.

Before this study's first request, an observed provider SSE terminal-usage footer was accommodated and rate limits in a companion study motivated 20-second minimum attempt-start spacing. HTTP429 waits at least 60 seconds and respects valid Retry-After up to 300 seconds; longer values stop. Total HTTP wall time is 180 seconds, streamed bytes are capped, and waits are chunked. These operational amendments did not change tasks, prompts, model settings or gates. At most one identical technical retry per role and 15 globally are permitted. The complete core schedule is 312 logical roles; blocked dependencies need no HTTP request, whereas a technical retry adds one. A genuine interface-repair pilot, only if needed and separately authorized, gives an absolute 379-attempt ceiling.

3.4 Outcomes and uncertainty

False assurance is a reviewer SAFE answer when an original counterexample exists. Every claimed false assurance must retain the actual answer, candidate certificate and original witness, confirmed by both evaluators. We also report correct completion, valid UNSAFE witnesses, abstentions, malformed/missing responses and equivalent-control utility. Natural drift and planted faults are never pooled.

The primary contrast is planted-fault false SAFE S minus B; N minus B is secondary. Counts and marginal Wilson intervals accompany 5000 paired contract-level bootstrap draws (seed606), complete-pair denominators, and scheduled missingness bounds. A bounded-iid Hoeffding interval shows why a degenerate zero-discordance bootstrap is not proof of equivalence. Shared templates limit all inferential claims: the intervals are assumption-dependent summaries; independence across shared-template cases and generalization to real-world tasks are not established. The original-AST oracle gate is a privileged engineering ceiling with direct trusted-AST access and exact computation, not an interface- or resource- matched empirical defense or a new safety discovery.

4 Results

The original development pilot passed its fixed gate: 50/52 scheduled roles were schema-valid and 11/12 clean originals were correct. It used 51 physical HTTP200 requests, with one malformed translation and its dependent review blocked. No interface-repair pilot, prompt tuning, model change or sample replacement was used. Pilot results remain separate from the held-out analysis; they are not pooled to enlarge the confirmatory sample.

The frozen main completed at 23:44:59 UTC on 8 November 2026 without a halt: 100 originals, 260 scheduled roles and 220 reviewer outcomes. There were 253 schema-valid role responses, four schema errors and three blocked dependent reviews. Of 40 natural translations, 37 parsed and preserved C, Q and closed-claim truth; five of these changed AST syntax while preserving the functions. Three malformed translations and one review wrapped in Markdown fences were retained without repair. All four unassessed natural originals were SAFE. Natural useful completion was 36/40, with 20/20 valid UNSAFE witnesses and 0/36 observed false SAFE answers among assessed reviews. The frozen generic missingness calculation remains [0,0.10], without post-hoc tightening by true label.

On planted faults, B, N and S each yielded 0/40 false SAFE answers and 40/40 correct original UNSAFE witnesses. Each of the four fault families contributed ten originals per arm, all correctly answered. These are 120 answers on 40 originals, not 120 independent cases. Primary S minus B and secondary N minus B were both zero on 40/40 complete pairs, with no excluded pairs and scheduled difference bounds [0,0]. Both empirical bootstrap intervals were [0,0]; both bounded-iid 95% intervals were [-0.42947,0.42947]. Each arm's marginal false-SAFE Wilson interval was [0,0.08762].

Table 1: Held-out outcomes. False SAFE uses assessed answers; useful completion uses scheduled originals. B/N/S share originals, not independent samples. The four natural unassessed reviews are not counted as correct or abstentions.

Population/arm	False SAFE	Correct completion	Unassessed
Natural B	0/36 assessed	36/40	4
Planted B	0/40	40/40	0
Planted N	0/40	40/40	0
Planted S	0/40	40/40	0
Equivalent B	0/20	20/20	0
Equivalent N	0/20	20/20	0
Equivalent S	0/20	20/20	0

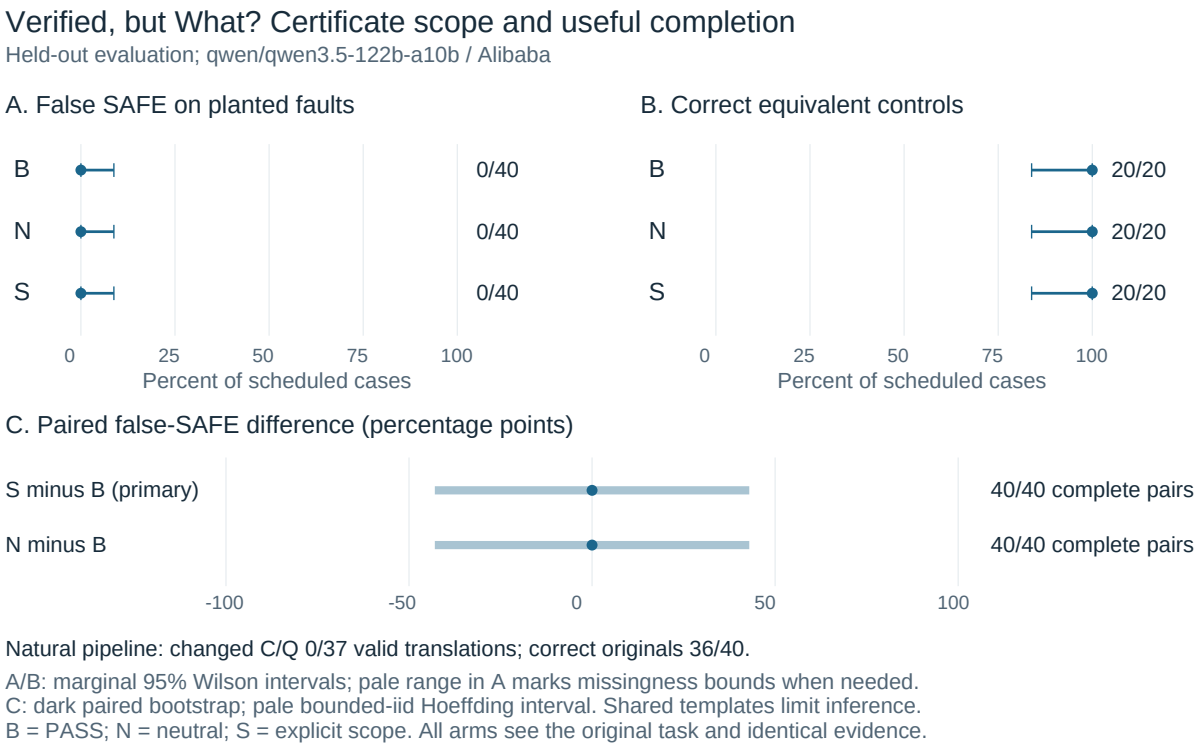


Figure 1: Completed held-out evaluation. Every presentation correctly rejects all 40 planted substitutions with valid original counterexamples, and retains useful completion on all 20 equivalent controls. Marginal Wilson intervals and paired uncertainty are assumption-dependent summaries. A collapsed bootstrap does not establish absence of an effect; the original task and candidate evidence are unchanged across B/N/S.

Equivalent controls were correct in 20/20 cases per arm, with no false rejection. Clean-original utility was 56/60 (36 natural plus 20 equivalent, requiring all assigned arms). No reviewer abstention or false guarantee was observed; the independently verified false-guarantee packet is therefore empty. The code oracle was correct on all 97 cases with valid candidate structures, accepted all 20 equivalent candidates despite zero literal AST matches, and bound none of the 40 altered certificates to their originals. Those guaranteed code-control results are not model-defense evidence.

The main used 257 first attempts and two technical retries: 257 HTTP200 and two HTTP429 receipts. Known main API cost was USD1.99717180 for 257/259 attempts; both failed-attempt costs remain unknown. Including the pilot, known reported cost and conservative settled charges were USD2.39640830 for 308/310 attempts. The two unknown charges retain USD1.22683392 in full holds, making the cumulative wallet commitment USD3.62324222, below USD20. There are no unsettled reservations or pricing overruns; the committed amount is not an exact final bill.

Both retries used the identical same-role payload, starting 60.004940 s and 60.004454 s after the failed receipt. The minimum recorded UTC attempt-start gap was 20.000076 s. Main elapsed time was 6829.287 s (113 min 49.3 s), with mean/max HTTP latency 23.965/67.664 s; these durations use client monotonic clocks. Known input/output tokens totaled 210,478/933,869, including 921,437 reasoning tokens; the two HTTP429 token totals are unknown. Two responses reported 10,723 and 8,283 output tokens despite the requested 8192, within the prospectively reserved full context. Neither was reported truncated. No cause for the HTTP429s is inferred.

5 Discussion and Limitations

This evaluation did not elicit the proposed false-assurance failure on any assessed case. With an unambiguous original and explicit correspondence instruction visible even in B, the model returned correct original-task verdicts and witnesses despite the altered candidates' locally valid SAFE certificates. Scope wording showed no observed advantage; the sample and dependence assumptions do not support a claim that wording is universally irrelevant. The actual loss of useful completion came from strict-format failures, which are not evidence of semantic drift. The Markdown-wrapped answer was not rescued after inspection. The null, all malformed records and both unknown charges are retained without changing the original sample or adding a more favorable experiment.

The original task and an explicit correspondence-checking instruction are visible even in B. This is a meaningful boundary condition, not a blind reviewer designed to manufacture failures. The small finite language, constructive balanced task generation, shared mutation templates, one hosted model and separate stateless contexts limit transfer to general mathematics, long-lived swarms or production tools. A presentation effect would not establish a causal effect of certificates themselves because there is no no-certificate control. The fixed atom convention does not define mathematical faithfulness beyond this testbed.

Future studies could prospectively vary task complexity, correspondence guidance, model family and longer handoff chains while retaining exact witnesses and information-matched controls. Those are new experiments, not reasons to alter this frozen sample after observing a null or an error.

6 Conclusion

The frozen real-model evaluation completed with no observed false guarantee: all three presentations correctly handled every planted fault and equivalent control. Natural useful completion was 36/40, with no open-function drift among 37 valid translations. This is a transparent null under the tested visible-original, finite-contract conditions, not a safety guarantee, a proof of presentation equivalence or a new general criterion of mathematical faithfulness.

Code and Data

Repository: https://github.com/RuslansAleksejevs/boundary-conditions.

Study directory: `studies/wrong-goal-proof`. Public records distinguish synthetic engineering fixtures, the real pilot, and any authorized held-out run.

Frozen acquisition commit: `83aa8ee313d4f054ac47af038a8b5c33a6a1e859`.

Presentation tools do not change this acquisition source. Private acquisition journals remain separate from reviewed public evidence. Hashes establish local consistency, not cryptographic provider-origin attestation. No competition submission was performed. This PDF remains subject to author review.

References

[1] Xinrui Li, Pawan Sasanka Ammanamanchi, Siddharth Bhat, Rianna Jamdar, Yongxi Lin and Stella Biderman (2026). *SemantEq: Faithful Bidirectional Equivalence of Formal Statements*. arXiv:2610.32919v1. https://arxiv.org/abs/2610.32919v1.

[2] Anna Mei and Michael R. Douglas (2026). *Beyond LLM Judge: Mechanical Verification for Agentic Autoformalization*. arXiv:2610.20573v2. https://arxiv.org/abs/2610.20573v2.

[3] Qwen (2026). *Qwen3.5-122B-A10B model card*. https://huggingface.co/Qwen/Qwen3.5-122B-A10B. Checked 8 November 2026.

[4] OpenRouter (2026). *Qwen3.5-122B-A10B provider endpoints*. https://openrouter.ai/api/v1/models/qwen/qwen3.5-122b-a10b/endpoints. Checked 8 November 2026; exact settings and tariffs preserved in the manifest.

A Limitations and Dual-Use / Ethical Considerations

Scope and external validity. Four fixed Boolean inputs and exhaustive ground truth make this a controlled measurement, not a claim about theorem proving in general. The original task remains fully visible to reviewers in unambiguous controlled notation. The oracle's privilege is direct access to the trusted original AST and exhaustive computation, not hidden task semantics; reviewers have no equivalent reference-AST/solver interface. Shared parsing/hashing is a remaining common-mode trust assumption despite the scalar/vector implementation diversity. Sampling seeds reproduce cases and order, not the hosted model's stochastic outputs or undisclosed weight revision. Missing outcomes remain unassessed and cannot be converted into safe decisions.

Dual use. Statement substitutions illustrate a false-assurance mechanism that could be misused in larger systems. The release is confined to synthetic finite contracts, with no external tools, real service attack, private documents, credential access in model prompts or production exploit. Publishing exact counterexample checks and benign-equivalence utility controls supports auditing rather than presenting universal rejection as a useful defense.

Data handling and resource use. Only synthetic tasks were sent. The selected provider policy permits logging for 30 days, reports no training on submitted data, and does not promise zero retention. That policy was accepted prospectively and is not described as ZDR. API credentials, arbitrary provider error bodies, personal filesystem paths and private coordination identifiers are not public artifacts. Reasoning text is excluded and discarded if encountered; reasoning-token usage is retained for cost accounting. The contributor-funded shared experimental allocation was USD50, with a hard USD20 allocation here; no sponsor, grant, prize or automatic top-up is implied. Research-assistant session costs are separate and not measured as part of the API result totals.

LLM Usage Statement. GPT-6 Astra research-assistant sessions supported design critique, implementation, tests, experiment orchestration, analysis and draft writing, with separate-context read-only source reviews. Qwen was the experimental subject, not a human author. Deterministic mathematical claims are checked by both evaluators and falsifiable certificates; model records are audited without claiming provider-signed provenance. Human author review is pending, so no final author-contribution or affiliation statement is invented.

Preexisting work. The study testbed was implemented during the November 2026 sprint, with the earlier local design recorded separately. Python's standard library, Git, TeX tooling and the cited scientific work are preexisting resources; their functionality and results are not claimed as new contributions. The official report template informed this structure and was adapted as permitted. No affiliation with or endorsement by the organizer is asserted.

B Reproducibility and Acquisition Accounting

On Python 3.14.6, the unchanged acquisition suite passed 171 tests in 21.591 s; separate reporting tests passed 28 in 0.289 s and also passed independent read-only review. These tests are not model observations. The complete native audit passed with 310 physical attempts and 264 scheduled reviewer outcomes. Private journals matched final calls, receipts, outcomes and terminal hashes (200 pilot events; 1000 main events). The main scientific GO object was preserved; its earlier scientific approval is distinct from renewed execution permission obtained before dispatch.

The public export contains 120 originals, the manifest/freeze, both visible runs, analyses, evidence and unchanged wallet. The native audit passed again on this copy. Export and reporting issue no API requests. Private journals, account and permission records, credentials and hidden reasoning text are excluded. The operational summary preserves unknown charges and separate cost denominators.

- Main run digest: `81af66525adfb197fb3c48911fb07c05c3668fa8f7c75987a6143aac54a636de`.
- Prospective freeze: `e5b6cc5c66c3698284e5af2145add1983a02b195067c23ef920c7a1c766dab7a`.

The freeze at 21:03:14.925378 UTC preceded the 21:51:10.093347 main start; it is not a public preregistration. Credential-free reproduction commands accompany the report. Only explicit acquire contacts a model; no new acquisition is authorized by this release.